

Quantum Deep Learning Still Needs a Quantum Leap

Hans Gundlach¹ Hrvoje Kukina² Jayson Lynch¹ Neil Thompson¹¹MIT FutureTech, CSAIL, Cambridge MA, USA | ²TU Wien, Vienna, Austria

OUR POSITION

Even accounting for rapid hardware progress, quantum computers will need a **quantum leap** — not just steady improvement — to meaningfully accelerate deep learning over the next **10–20 years**.

Why Ask Now?

Deep learning's progress tracks **compute** — the “bitter lesson.” As classical hardware nears its limits, it is natural to ask whether **quantum hardware** could drive the next paradigm shift, as GPUs once did.

We present a **first-of-its-kind survey** mapping quantum algorithms onto every stage of the deep-learning pipeline, paired with **quantitative economic-advantage forecasts** built on real quantum hardware trends.

The verdict: today's promising speedups are blocked by overhead, missing memory, or apply only to narrow special cases.

Quantum Economic Problem Size For a Given Year

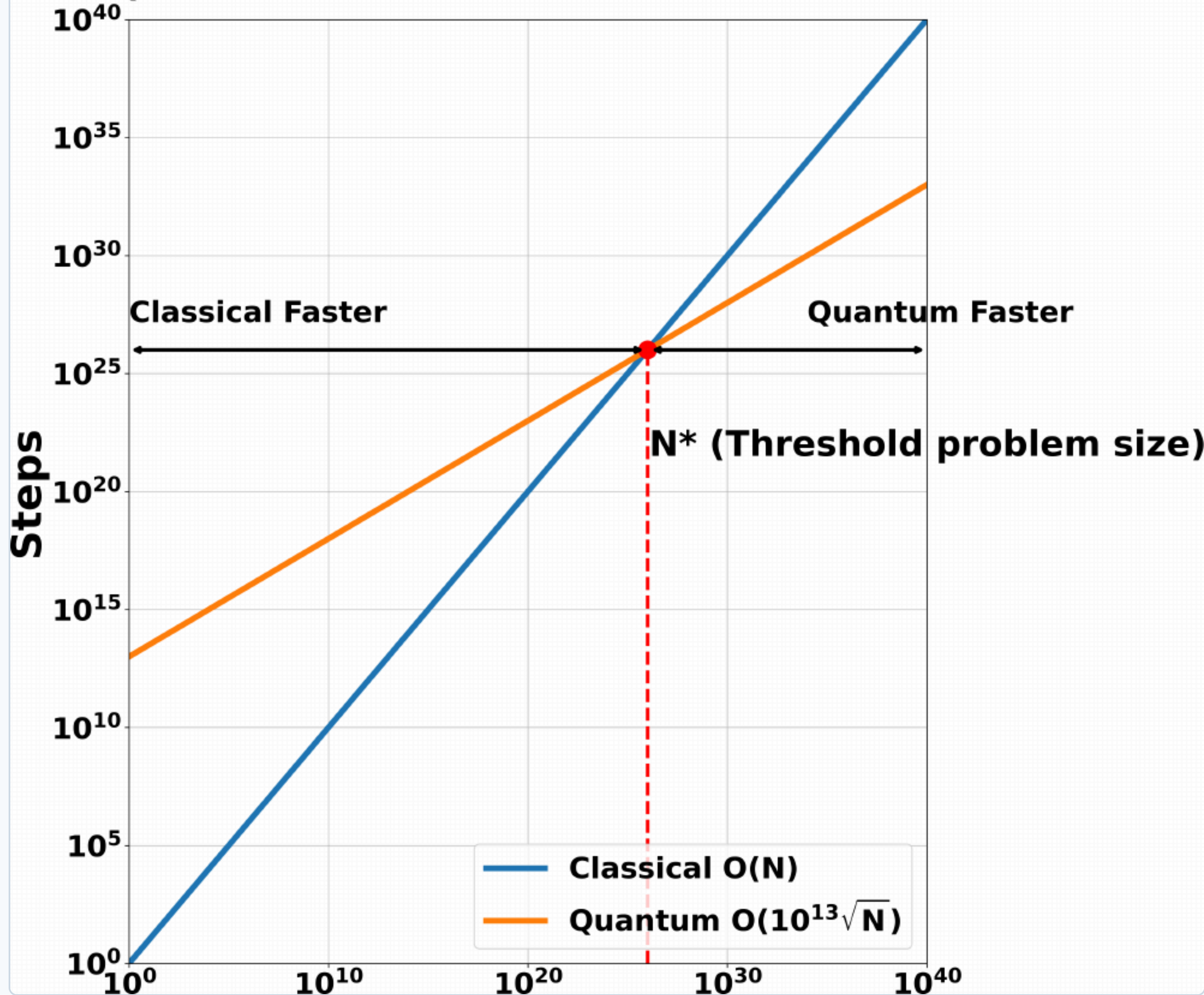


Fig 1 · Quantum Economic Advantage (QEA). Quantum gains are asymptotic but carry a huge constant hardware overhead. A problem pays off on quantum only beyond a critical threshold N^* — below it, classical is faster.

The QEA Model

We extend the Quantum Economic Advantage framework of [Choi et al. \(2023\)](#). Quantum devices have asymptotic advantages but enormous constants:

- $\sim 10^{13} \times$ slower gate speed vs. a floating-point op
- $\sim 10^2 \times$ overhead from quantum error correction
- $\sim 10^8 \times$ more GPU parallelism per dollar classically

We forecast how **qubit counts**, **gate speeds**, and **error rates** evolve, then compute when—if ever—each algorithm crosses its threshold.

Quantum vs Classical Search



Fig 2 · Quantum vs. classical search. Accounting for hardware trends and a 1-month runtime limit, Grover-type search reaches economic advantage only around **2034**, and only for problem sizes far larger than real deep-learning search.

Where Could Quantum Help?

We assess the best available quantum option at each stage of the deep-learning pipeline — and its status today.

DL STAGE	BEST QUANTUM OPTION	SPEEDUP	STATUS
Data processing	q-means; quantum PCA	Exponential	Blocked by QRAM
Hyperparameter & action search	Grover; Dürr-Hoyer	Quadratic	Infeasible ($N > 10^{24}$)
General matmul / attention	Swap-test; Gao attention	Polynomial	Lost to overhead; QRAM limits
NN training (classical)	Allcock; Liu; Newton; Zlokapa	Mixed	Promising, but slow / restrictive
Quantum NNs / VQAs	VQAs; kernel methods	Unclear	No advantage in benchmarks

Table 1. Status of quantum acceleration across the deep-learning pipeline.

Robustness

We vary every growth rate and constant in the model across 10^{-3} – 10^4 . The predicted advantage year is **largely insensitive** to these changes.

The one real lever is the **quantum gate-speed improvement rate** (beyond Moore's law), which materially shifts the outlook for matrix multiplication. Grover predictions are robust — these algorithms are **time-bound, not qubit-bound**.

Quantum Economic Advantage Analysis by Algorithm Type

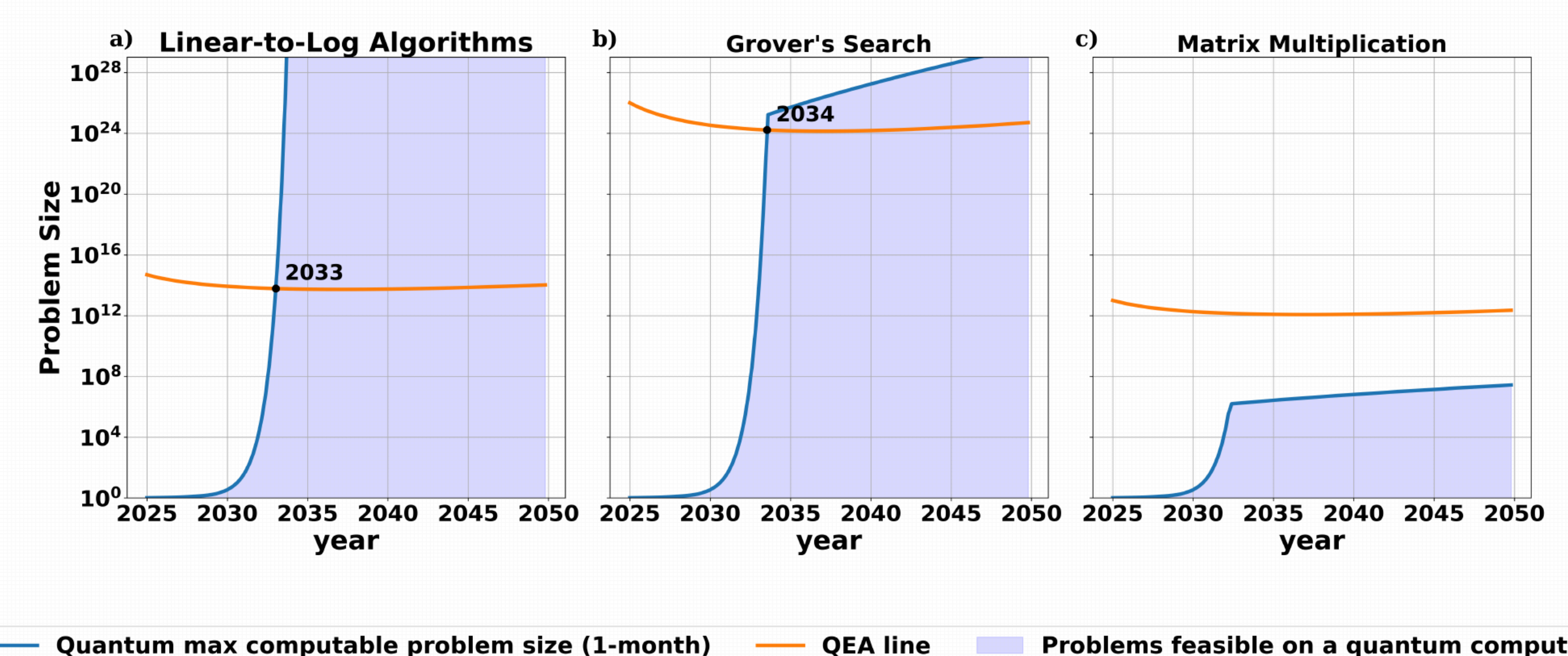


Fig 3 · Advantage by algorithm class. Problem sizes in the shaded region are feasible on a quantum computer; only those above the orange QEA line benefit. (a) exponential speedups can win by ~2033; (b) Grover wins only at enormous sizes (~2034); (c) dense matrix multiplication stays out of reach.

3 Three Roadblocks

- 1 Overhead swamps matmul.** Quantum matrix multiplication beats classical $O(N^3)$ asymptotically, but constant overhead makes it infeasible at any practical size for foreseeable hardware.
- 2 QRAM is missing.** The most powerful speedups — q-means, quantum PCA, HHL-based training, kernel methods — all depend on practical **QRAM**, which remains underdeveloped.
- 3 Special cases only.** Grover-type quadratic speedups need search sizes $> 10^{24}$ to pay off — far beyond real hyperparameter or RL action search.

Call to Action

- **Accelerate QRAM** — it gates most of the promising exponential speedups.
- **Push quantum gate speed** and cut error-correction overhead beyond current roadmaps.
- **Target problems** with both a clear asymptotic advantage **and** viable problem sizes.
- **Stay skeptical** of exponential “training” claims that rest on heavy preconditions.

Quantum's near-term role in deep learning is **narrow**. The opportunity is real — but realizing it requires a leap, not a step.